Lesson Notes

MODULE 3 | LESSON 1

THE BLACK-LITTERMAN MODEL

Reading Time	90 minutes
Prior Knowledge	Traditional mean-variance optimization, Working knowledge of Bayesian theory, Matrix algebra
Keywords	CAPM, reverse optimization, equilibrium excess returns, market portfolios and global market portfolio, Black-Litterman model.

The previous lesson centered around factor models. Factor models can be seen as an alternative to more traditional mean-variance optimization in the sense that they allow an investor or a portfolio manager to select their desired exposure to specific asset classes.

In this lesson mean-variance optimization is brought back to the center stage, but with a new twist. Investors are allowed both to have no views or to have strong opinions about how assets can perform and their views/opinions are allowed to enter the model and become a part of it. Incidentally, doing so also alleviates some of the issues that are least desirable about traditional mean-variance optimal portfolios (concentrated positions and instability of solutions to changes in inputs).

The next lesson will introduce the critical-line algorithm, which is an important tool both for traditional portfolio optimization and the Black-Litterman model.

1. What is the Black-Litterman Model About?

The Black-Litterman model is not a completely new technique, and it builds upon the same type of mathematical tools used in mean-variance optimization of portfolios. The novelty of the model is that it calculates expected returns using two sources of information:

- By estimating implied expected excess returns in equilibrium, the model is based on the current market's views.
- By allowing a portfolio manager to express one's views about excess returns of stocks, asset classes, or countries, the model incorporates the managers' experience and subjective views.

Putting together the two sources of information, the model produces new estimates of expected excess returns, which become the new inputs of a mean-variance optimization model. Despite the fact that there is a certain amount of pure theory behind it, the Black-Litterman model is very much an applied technique of portfolio theory. In fact, this model was created by experienced practitioners Fischer Black and Robert Litterman of Goldman, Sachs & Company in the very early 1990s.

The model also makes it possible to overcome the particular problem (that we saw in the previous lessons) of highly concentrated, input-sensitive portfolios.

Highly concentrated positions are a problem because financial theory preaches that diversification is a good thing and input sensitivity is bad news for portfolio managers. This is because portfolio managers may find themselves forced to undertake heavy rebalancing and/or end up with portfolios that are counterintuitive and/or containing massive long and short positions and therefore untradable.

Best and Grauer in a 1991 paper have shown that a small increase in the expected return of one of the portfolio's assets can force half of the assets from the portfolio. Some researchers have tried to explore alternative estimates for expected returns (historical, returns, equal-mean returns, risk-adjusted equal mean returns, and more), but no matter what they tried, the end results were portfolios with large long and short positions concentrated on a small number of assets.

The Black-Litterman model uses a Bayesian approach in which the investors' subjective views regarding the expected returns of one more assets are combined with the market equilibrium vector (the prior distribution) of expected returns. The result is a

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new mixed estimate of expected returns (the posterior distribution), which can be described as a weighted average of the investor's views and the market equilibrium.

Academic and applied research confirm that the methodology is largely successful in its ability to avoid the tendency to generate non-tradable, extreme, and oversensitive portfolios every time the parameters of the model are updated.

This lesson is based on the required reading "The Black Litterman Model in Detail".The assigned reading is a fairly detailed survey paper that contains the theory and the technical proofs (discussed in a mathematical appendix).To avoid the risk of "missing the forest for the trees", this lesson aimes at providing a narrative to simplify and/or justify certain aspects of the theory to guide students to read the assigned materials and reflect on a certain amount of details and best practices to implement the model in real case studies.

2. Market Equilibrium Portfolio and Reverse Optimization

2.1 Market Equilibrium

It should be known from previous essons that we can write the linear regression model to test whether the CAPM holds as::

$$r_{it} - r_f = \alpha_i + \beta_i(r_{mt} - r_f) + \epsilon_{it},$$

where

- r_f is the risk-free rate;
- r_{it} is the return of asset i at time t;
- r_{mt} is the return of the equilibrium market portfolio at time t;
- ϵ_{it} represents the idiosyncratic risk for asset i at time t.

If $\alpha_i = 0$ then the CAPM model holds. The equilibrium market portfolio is the result of the decision making of all market participants, and it assumes that all economic agents hold symmetrical information about stock or asset classes returns, variances, and covariances. Furthermore, the model assumes that there is continuous and frictionless trading that allows an adjustment towards a long-term equilibrium. As such, the equilibrium market portfolio represents the shared objective view about excess returns (indirectly, the student should have recognized that we are postulating a semi-strong form of market efficiency). The idiosyncratic risk can be reduced or even made to disappear via diversification, and as such, it is not rewarded; only the systematic risk (measured via β) is.

We are also aware from a previous lesson that the Sharpe index for the market portfolio:

$$\frac{E[r_m] - r_f}{\sigma_m}$$

is also the slope of the capital allocation line (CAL) line, which allows us to identify the optimal (tangent) portfolio.

2.2 Reverse Optimization

The concept of reverse optimization was invented by Sharpe in 1974, and it is nothing too complicated, although the name could lead one to think that we are about to get involved with some type of sophisticated mathematical tool. Reverse optimization is simply used to derive the equilibrium expected excess returns from the CAPM via a certain number of parameters. To this purpose, we notice that the return on the market portfolio in equilibrium can be written in terms of the return of its components as:

$$r_m = \sum_{j=1}^{N} w_{m,j} r_j.$$

Now, let $\pi_i = E[r_i] - r_f$. Then, from the CAPM, we have

$$\pi_i = \beta_i(E[r_m] - r_f).$$

However, recalling that $\beta_i = \frac{\text{cov}(r_i, r_m)}{\sigma_m^2}$, we can rewrite the previous equality as

$$\pi_i = \frac{cov(r_i, r_m)}{\sigma_m^2}(E[r_m] - r_f) = \frac{cov(r_i, r_m)}{\sigma_m}\frac{E[r_m] - r_f}{\sigma_m} = \frac{cov(r_i, r_m)}{\sigma_m} \cdot SR_m, \text{ or}$$

$$\pi_i = \frac{E[r_m] - r_f}{\sigma_m^2} \sum_{j=1}^{N} \text{cov}(r_i, r_j) w_{m,j},$$

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where we just used the definition of r_m and the linearity property of the covariance operator.

Finally, if we let

$$\Sigma = \begin{pmatrix} \operatorname{cov}(r_1, r_1) & \operatorname{cov}(r_1, r_2) & \cdots & \operatorname{cov}(r_1, r_N) \\ \operatorname{cov}(r_2, r_1) & \operatorname{cov}(r_2, r_2) & \cdots & \operatorname{cov}(r_2, r_N) \\ \vdots & \vdots & \ddots & \vdots \\ \operatorname{cov}(r_N, r_1) & \operatorname{cov}(r_N, r_2) & \cdots & \operatorname{cov}(r_N, r_N), \end{pmatrix}$$

then we can write in matrix notation:

$$\pi = d\Sigma w_m$$

with π the $N \times 1$ vector of risk premiums in equilibrium,d is the risk-aversion coefficient, which corresponds to the Sharpe ratio, Σ is the $N \times N$ matrix of excess returns for the N assets, and w_m is the $N \times 1$ vector, which represents the weights of the individual assets in the portfolio based on market capitalization.

Why is it called "reverse optimization"? Following Sharpe in a 2007 paper, we could describe mean-variance portfolio optimization as a process like:

$$\begin{aligned} &\textbf{expected returns} \ + \ \textbf{risk} \ + \ \textbf{correlations} \\ &+ \ \textbf{risk aversion coefficient} \ \mapsto \ \textbf{optimal portfolio.} \end{aligned}$$

Using the market portfolio as the optimal portfolio is a legitimate choice if we assume that all economic agents have the same information and the same coefficient of risk aversion because they would all compute the same optimal portfolio. Then, the previous equation can be rewritten as:

$$\begin{aligned} &\textbf{market portfolio} \ + \ \textbf{risk} \ + \ \textbf{correlations} \\ &+ \ \textbf{risk aversion coefficient} \ \mapsto \ \textbf{expected returns.} \end{aligned}$$

Note that the risk-aversion coefficient,d in our notation, acts like a scaling factor for the reverse optimization estimate of excess returns. In the next section we will show that it corresponds to the Sharpe ratio for the market portfolio.

An individual who has no personal views about the market should hold the market portfolio, which becomes a reference point. In the language of Bayesian statistics, it could also be referred to as an **uninformative uniform prior.**

Arguably, the most tedious part about the Black-Litterman model is the preparation of the inputs as we need to identify the investment universe and estimate excess returns, covariance matrices, and risk-aversion coefficients. A global index (e.g., the MSCI World Index, UBS's GSMI, etc.) can be used as proxy for the market portfolio.

Since the CAPM contains all investable assets, it is very hard to specify.

However, because we are considering a situation of equilibrium, all sub-markets must also be in a situation of equilibrium; therefore, we could choose any investable sub-universe to act as the market portfolio.

Important caveat: If a portfolio manager decides that their universe of possible investments is just a handful of stocks, that would become the market portfolio. Because of our remark in the previous paragraph, that is legitimate. This makes computing the weights of the assets in the market portfolio much simpler. However, we need to be careful that now we need to compute the excess returns and the Sharpe ratio for this new market portfolio we selected and not keep using the MSCI or the GSMI indexes! In other words, now we need to construct the excess returns for our own market portfolio, which acts as our unique benchmark. So we gain in one direction, but we lose in another as now we need to construct the returns for our unique market portfolio weighing the returns of each of the component assets.

Example (Equivalence with approach based on implied betas). An alternative choice is to use the market portfolio capitalization weights to compute expected excess returns—that is, we could use what are called **implied betas**. What are implied betas? Implied betas are the betas of the N assets relative to the market capitalization-weighted portfolio. The formula (for the full $N \times 1$ vector of betas) is given by

$$\beta = (w_m^T \Sigma w_m)^{-1} \Sigma w_m.$$

Once we have the vector of implied betas, we can find the vector of CAPM excess returns. It is possible to prove that the vector of CAPM excess returns coincides with the reverse optimized returns when the CAPM returns are based on implied betas.

Why do we bother so much with market portfolios and the reverse optimization to produce excess equilibrium returns? Well, it turns out that these estimates are less volatile then using historical ones and produce asset weights that are much less extreme

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besides allowing the portfolio manager not to deal with very large long and short positions. As these are also inputs required for the Black-Litterman model, this is important because the methodology leads to cleaner and less extreme portfolios.

Example To show how to compute the excess returns in equilibrium we are going to use an example in which the market portfolio is the S&P 500 Index. As the individual assets we are considering the 11 sectors of the index listed immediately below with their tickers:

S&P 500 SECTORS

^SP500-15 Materials sector
^SP500-20 Industrial sector
^SP500-25 Consumer discretionary sector
^SP500-30 Consumer Staples sector
^SP500-35 Healthcare sector
^SP500-40 Financials sector
^SP500-45 Information Technology sector
^SP500-50 Telecommunication Services sector
^SP500-55 Utilities sector
^SP500-60 Real Estate sector
^GSPE Energy sector.

We are considering the period from January 1, 2022 to August 4, 2023 as our reference period. As of July 12, 2023, the weights of each of the 11 sectors with respect to the S&P 500 were in terms of capitalizations (with minimal roundings so that they sum to 1):

^SP500-15 0.02
^SP500-20 0.08
^SP500-25 0.11
^SP500-30 0.075
^SP500-35 0.135
^SP500-40 0.13
^SP500-45 0.275
^SP500-50 0.09
^SP500-55 0.02
^SP500-60 0.02
^GSPE 0.045

Furthermore during the period $\sigma_m = \sigma_{SP500} = 0.21$ and the Sharpe ratio for the S&P 500 was calculated at 0.62. These numbers are likely to change over different periods, so it is important to specify the time interval used.

At this point we will use some code to compute the excess return in equilibrium, π which will become one of the inputs later when using the Black-Litterman approach.

```
#---------------------------------------------------

import libraries (some are for other applications)

#---------------------------------------------------
import yfinance as yf
import datetime
import math
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
#---------------------------------------------------

uire the data using Yahoo Finance
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sectors=["^SP500-15", "^SP500-20","^SP500-25","^SP500-30","^SP500-35",
"^SP500-40","^SP500-45","^SP500-50","^SP500-55", "^SP500-60",
"^GSPE" ]
start = datetime.datetime(2022, 1, 1)
end = datetime.datetime(2023, 8, 4)
data = yf.download(sectors, start=start, end=end)['Adj Close']

plot the time series for the individual sectors

data.plot.line(y=sectors, figsize=(10,6),title='S&P500 sectors', ylabel='level' )

calculate the geometric returns

df = np.log(data).diff()
df1= df.dropna()
#-----------------------------------------------------

Compute annualized mean returns and covariance matrix

#-----------------------------------------------------
def expected_return(dataframe):
expected_returns = []
for i in df1.columns:
expected_returns.append(df[i].mean()*252)
return np.array(expected_returns)
returns = expected_return(df1)
df_cov = df1.cov()*252
df1_cor = df1.corr()

-------------------------------------------------------------------------------

weights for each of the sector (manually coded as Yahoo Finance does not

provide this type of data)

#-------------------------------------------------------------------------------
weights_sectors = [0.02, 0.08, 0.11, 0.075, 0.135, 0.13, 0.275, 0.09, 0.02,0.02,0.045
d = 0.62/0.21  # d=2.95
d*(df_cov.dot(weights_sectors))
```

This produces the following excess returns in equilibrium, π **:**

^GSPE 0.083615

^SP500-15 0.115286

^SP500-20 0.104200

^SP500-25 0.145102

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^SP500-30 0.065655

^SP500-35 0.077166

^SP500-40 0.117381

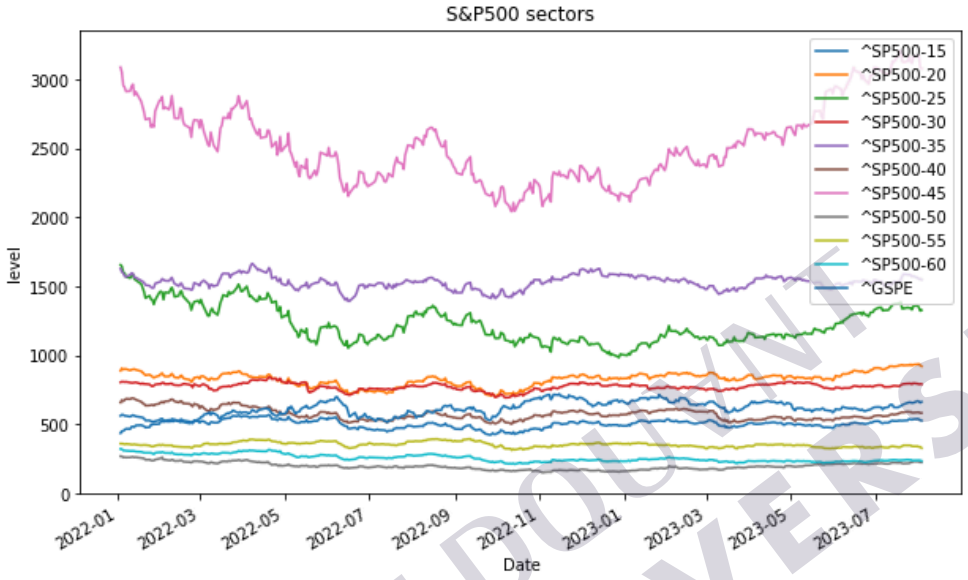
^SP500-45 0.141676

^SP500-50 0.127710

^SP500-55 0.070367

^SP500-60 0.105803

Figure 1: The time series for the 11 sectors that compose the S\&P 500 Index during the period from January 1, 2022 to August 4, 2023.



3. Black-Litterman Model: Implementation

The Black-Litterman model starts from the market equilibrium and introduces in it the portfolio manager's opinions about the returns for each asset. The model does so using a Bayesian approach:

- investor's views play the role of prior distribution;
- market excess returns in equilibrium play the role of additional information;
- Bayes' theorem produces optimal weights like a posterior distribution.

In what follows, we explain the Black-Litterman model using steps to break down the procedure as described in the assigned reading. Some are straightforward while one or two require a bit more thinking.

Take your time to understand each of the steps.

Step 1: The expected returns are assumed to follow a normal distribution, i.e.,

$$E[r] \sim N(\mu, \Sigma).$$

However, μ is not a parameter but a random variable. We express this as:

$$\mu \sim N(\pi, \Delta)$$

in which π is an estimate of the mean μ and Δ is the variance of the estimate μ. The relationship between μ and π can be written as

$$\mu = \pi + \epsilon$$

or

$$\pi = \mu - \epsilon.$$

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If $\epsilon \sim N(0, \Delta)$, then $-\epsilon \sim N(0, \Delta)$ as well, and we can just write

$$\pi = \mu + \epsilon.$$

If we make the further assumption that μ and ϵ are not correlated, then simple properties of the normal distribution yield:

$$E[r] \sim N(\pi, \Psi \equiv \Sigma + \Delta).$$

Before moving on to the next steps, we stress the importance of noticing the difference between the two formulations.:

$$E[r] \sim N(\mu, \Sigma) \quad \text{and} \quad E[r] \sim N(\pi, \Sigma + \Delta).$$

The second form is what is referred to as the **canonical Black-Litterman model**. However, some authors use the first model, and for this reason, it is always very important to be aware of what model is being used.

Step 2: In this step we need to specify π. Black and Litterman do this using the reverse optimization method seen earlier based on the CAPM and the market equilibrium, that is:

$$\pi = d\Sigma w_m$$

where the symbols have been introduced in the previous section.

Step 3: We need to calculate the coefficient of market risk aversion,d. To do that, we notice that if we start from the formula for π obtained in Step 2 and multiply both sides of the formula on the left by w_m^T, we obtain

$$w_m^T \pi = d w_m^T \Sigma w_m$$

(remember that d is a scalar quantity and as such can be moved anywhere in the formula). Now, we pay attention to the fact that

$$w_m^T \Sigma w_m = \sigma_m^2,$$

the variance of the market portfolio. Furthermore, we have also

$$\pi_m = E[r_m] - r_f = \sum_{i=1}^{N} w_{m,i} \pi_i \equiv w_m^T \pi$$

and therefore we have shown that

$$d = \frac{E[r_n] - r_f}{\sigma_m^2} = \frac{SR_m}{\sigma_m}$$

In other words d is the Sharpe ratio of the market portfolio divided by the volatility of the market portfolio which must be obtained from one of the global indices mentioned earlier.

Step 4: This is just a reminder that the weights w_{m_i} of each asset in the market portfolio are determined as

$$w_{m,i} = \frac{c_i}{c_m}$$

with c_i and c_m the capitalized value of asset i and the market portfolio, respectively.

Step 5: We need now to estimate Δ, the variance of μ and π. Black and Litterman make a simplifying assumption, i.e., they assume that

$$\Delta = \tau \Sigma$$

i.e.,Δ is proportional to the variance-covariance matrix of returns, Σ. In other words, they assume that

$$\mu \sim N(\pi, \tau \Sigma).$$

We eliminated one unknown quantity, but at the price of introducing another one. However τ is a scalar while Δ was a matrix. Hence, we have certainly simplified our problem.τ is a number between 0 and 1. The justification for this further assumption is that it measures the uncertainty of a mean. As such, its variability should be smaller than that of the individual excess return components.

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There is also another way to look at it: f an economic agent entertains the view that excess returns are very close to those calculated from the market portfolio via reverse optimization, then the variance of μ should be very close to zero. The value of τ can be calculated via a calibration process, but practitioners often assume that τ is a value in the range between 0.01 and 0.05.

Using MLE, the value of τ can be estimated as

$$\hat{\tau}_{MLE} = \frac{1}{T}$$

with T the number of observations. However, the MLE is not unbiased, and an unbiased version of the estimate is

$$\tilde{\tau} = \frac{1}{T-N}$$

where N was defined earlier as the number of assets in the market portfolio. These are not the only choices, and others have been proposed in the literature.

Step 6: When we work with the Markowitz model, economic agents/investors are allowed to have views regarding the expected excess returns of the assets. However, their views are absolute, meaning that they agree that an asset would achieve a return of, say, 3% more than the year before. In case they had no views, they would be assuming the same excess returns implied by the market portfolio via reverse optimization. In particular, the more a portfolio manager's views differ from the market portfolio's, the more extreme the shift would be in the weights of the individual assets vs. the weights in the market portfolio. No consideration is paid to the fact that views could be subject to errors and thus have a random nature.

The Black-Litterman model does not take this away. However, it allows for relative views/opinions (like when a manager estimates that an asset would produce a return higher (or smaller) than that of another asset. And it introduces random errors in the setup. For instance, let's keep assuming that we have N assets and that we have K opinions. We could express this as:

$$P\mu = q + \epsilon^v$$

where

P is a $K \times N$ matrix whose rows contains the weight of each of the investor's individual views. We have K rows, i.e., as many as the views the investor entertains about the overall universe of views. In case an investor had no view about one or more of the assets, then the corresponding entry along the row will be zero.μ is the $N \times 1$ vector that contains the average of expected excess returns,q is a$K \times 1$ vector that contains the returns forecasted for each of the K views. Finally, ϵ^v is a $K \times 1$ vector that contains the random error associated with the views.

At this point, an example is definitely needed to understand how this works in practice.

Example (Views about assets in Black Litterman). Let's assume that we have $N = 5$ assets and $K = 3$ views. The three views are as follows:

- During the next period, asset A will achieve a return of 8%;
- Again during the next period, asset B will achieve a return of 3%;
- During the next period, asset D will register a return higher than that of A,B,C, and E by 2.5%.

In this particular case P is a 3×5 matrix that looks like

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1/4 & -1/4 & -1/4 & 1 & -1/4 \end{pmatrix};$$

q is a3×1 matrix that looks like

$$q = \begin{pmatrix} 0.08 \\ 0.03 \\ 0.025 \end{pmatrix}.$$

The errors in the views are accounted for with the assumption that\

$$\epsilon^v \sim N(0, \Omega)$$

with Ω a $K \times K$ matrix that incorporates the manager's uncertainty about each view. In particular, the views are independent fr ach other and therefore Ω is a diagonal matrix, that is $\Omega = \text{diag}(\omega_1^2, \ldots, \omega_K^2)$. In turn, this implies that

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$$P\mu \sim N(q, \Omega).$$

Ω is a critical input, and it depends on the individual investor as they are the best economic agent to decide the certainty of their opinions. However, in the literature, a few formulas have been proposed and one often used is the following:

$$\Omega = \text{diag}(P(\tau\Sigma)P^T).$$

A still different approach could start with the individual investors assigning a confidence level to each of their views. For instance, an investor could think that asset i will appreciate by 5% during the next period, and he is 95% confident that the return will be between 1% and 9%. In this case, using simple rules for normal random variables it is easy to realize that $\omega_i = 2\%$.

Example (Views and confidence about assets in Black Litterman). Assume that we have 4 assets (A, B, C, D) and an investor with the following views:

- Asset A will achieve a return of 5% more than asset C. He think that his confidence is $\omega_1 = 3\%$.
- Asset D will achieve a return of 3% more than in the previous period. Confidence level $\omega_2 = 4\%$.

Well, in this case we have

$$P = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad q = \begin{pmatrix} 0.05 \\ 0.03 \end{pmatrix}, \quad \Omega = \begin{pmatrix} .03^2 & 0 \\ 0 & 0.04^2 \end{pmatrix}.$$

Step 7: This is the complex step, but the good news is that while the calculations are hard (we need to compute the mean of a posterior distribution using a multivariate normal with a multivariate normal prior), the operational result is available. The posterior distribution for the expected excess return for the assets that incorporates the portfolio manager's views is given by

$$E[r] = [(\tau\Sigma)^{-1} + P^T\Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^T\Omega^{-1}q]$$

which can be shown to be equal to

$$\mu_{BL} := E[r] = \pi + \tau\Sigma P^T[(P\tau\Sigma P^T) + \Omega]^{-1}[q - P\pi].$$

This second expression has an advantage in the sense that one does not need to invert the matrix Ω, which can make computations more precise and/or faster. Keep in mind that usually the matrix Ω contains small entries along the main diagonal and, on inverting, there could be some numerical loss of precision.

The posterior covariance matrix is given by

$$\text{cov}(r) \equiv M = ((\tau\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1}.$$

Is important to keep in mind that the posterior covariance matrix just supplied is the covariance matrix of the posterior mean estimate about the actual mean, or, said differently, it measures the uncertainty in the estimated posterior mean. If we wanted to compute the variance of the returns to use in a mean-variance optimization program, we need to compute

$$\Sigma_{BL} := \widetilde{\Sigma} = \Sigma + M = \Sigma + ((\tau\Sigma)^{-1} + P^T\Omega^{-1}P)^{-1}.$$

Step 8: Now we are ready to perform the last step, i.e., we can run the following optimization program.

$$\begin{gathered} \max_w w^T\mu_{BL} - \tfrac{1}{2}w^T\Sigma_{BL}w \\ \textbf{subject to: add constraints, if any.} \end{gathered}$$

4. Further Readings

One could ask what is the outcome if the investor's views are wrong or only partially correct? Clearly, the result obtained using the Black-Litterman model would not be optimal. To be precise, it is optimal in the sense of the model, but it is not optimal from the point of view of practical results. Often, upon noticing that the portfolio is not performing well, the investor will correct their views and obtain a new optimal portfolio. However, to do so means that there will be rebalancing and associated transaction costs. Yanou in 2010 has proposed an alternative to the standard Black-Litterman model that incorporates the possibility of wrong views and opportunity costs. Yanou's model tries to arrive at an optimal solution when the opportunity costs are taken into consideration as part of the model itself.

Yanou's working paper can be found here: The Black-Litterman Model: Wrong Views vs. Opportunity Cost

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5. Python Implementation

As we said several times, nothing has really changed vs. the regular mean-variance optimization problem, except that we need to obtain some additional inputs and create views. A working Python implementation can be found here:

Stuart Jamieson, Black-Litterman Portfolio Allocation Model in Python (2020)

Another couple of useful links are the following, which points to the PyPortfolioOpt package that contains general tools for portfolio optimization and contains the Black-Litterman model, among other things:

- PyPortfolioOpt
- Robert Andrew Martin PyPortfolioOpt: portfolio optimization in Python, Journal of Open Source Software, 6, (61), 2021

6. Conclusion

In this lesson, we learned a different way to do mean-variance optimization to determine optimal portfolios. More specifically, we learned how to estimate an optimal portfolio using the consensus view (the market's view) and how to introduce the portfolio manager's views into the problem. The Black-Litterman model is a relevant tool for portfolio management, and it provides a generalization of previous algorithms we saw in earlier lessons.

In the next lesson, we will cover the critical line algorithm, which is quite useful in solving portfolio optimization problems (both for the traditional problem and Black-Litterman) when inequality constraints are required.

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